

PARALLAX — Master Idea Bank

Everything parked, consolidated. Retirement-planning simulator for advisors.

Generated 2026-06-03 · source: ROADMAP.md + NOTES.md + CLAUDE.md · tags: NEXT · BIG (own session) · PARKED · SKIP ·  done

1 · Engine — material gaps truth source

The single truth source. The UI never adds math — only views/levers of what the engine emits.

- **Real assets in engine** (homes / property / business) → "sell X to fund Y." BIG
- **Tax-planning engine** — Roth conversions, bracket management, IRMAA, RMDs, contribution-side deduction. Its own engine. BIG
- **Survivor SS benefits** — widow step-up (~100% of deceased's); major couples lever, unmodeled.
- **Healthcare scales independently of lifestyle** — spendMult currently scales medical too; skews Solve-For.
- **LTC cost escalation** — ~3–5%/yr above CPI; currently flat real.
- **One-time inflow support** — relocation / downsizing; current lumpSum is outflow-only.
- **Map the other engines** we'll need (tax, estate...) before building piecemeal.
- Cleanup: `ASSET_STATS[k].mean` computed but unused — deletable.

2 · Core differentiators — the moat big

The defensible layer: deterministic, explainable, multi-variable decision-solving — not "Monte Carlo + sliders."

- **Solve-For mode** — named life questions: ① "Can I retire earlier?" (build first) ② "Can we afford this big goal?" ③ (later) "Spend more?" On-demand (one engine job/click); output = a new scenario column; exactly one free lever. NEXT-ISH
- **Explainable constraint solver** — multi-variable search returning a small Pareto set of least-disruptive paths; auditable replay. Also fixes the solo-vs-combined display confusion.
- **Decision Surface heatmap** — retire-age × spend, confidence contours, saved scenarios as clickable points. The best embodiment of "Parallax models interactions." Alternate view in Scenarios.

3 · High-value views of existing truth low lift

Mostly a VIEW of data the sims already emit — no new math.

- **Failure Anatomy** — click a % ring → depletion-age histogram, earliest pressure year, success/depleted split, one representative failed path in the cash-flow drawer. **Best ROI.**
- **Scenario Receipt** — per-scenario delta drawer (levers moved + confidence/median before → after + worst market). Meeting takeaway / export. Data, not prose.
- **Sequence Pressure Strip** — first ~10 retirement years as cells (return / withdrawal / ending balance), cursor synced to chart + drawer. Explains *why* 1973 / 2000 hurt.
- **Funding Bridge / bridge-years** — stacked age timeline of income layers; the pre-SS/pension gap.
- **Honest derived tiles** — longevity buffer, max sustainable withdrawal %, "could retire at X."
- **Confidence-by-year bars** + a single "Confidence Age" — only if it's real per-year data.

4 · Scenarios / input model

- **Account-types picker** — "+ add investment account" with type picker (401k/403b/457/SEP/SIMPLE/solo-401k → pre-tax). MOCK BUILT, AWAITING OK
- **Scenarios "shared-track" redesign** — kill the 3-column repetition; one shared track per lever, dots split only on divergence. Approved-in-spirit. BIG
- **Pension claim-age analysis** — sub-mode: claim-age sweep (62 → 70) + breakeven.
- **Real-assets input** — Money Pro balance-details layout (pairs with engine real-assets).
- **"Duplicate scenario"** action (distinct from Add).

5 · Sequencing

- **Strategy Fork on one path** — fix ONE market, compare strategies on it. Your #1 advisor function. Decide Scenarios-vs-Sequencing home first (overlap).
- **Recovery Tunnel** — full years-underwater valley view.
- **Sequence Duration / damage-window labels** — "short shock" ('08–09, V-shaped) vs "long grind" ('66 / '73). A label, not new math.

6 · This session's adds 2026-06-03

- **Home sale event + cap-gains on sale** — proceeds = value – mortgage – cap-gains tax; needs the \$250k/\$500k-MFJ primary-residence exclusion + a *capitalImprovements* basis field. purchasePrice already captured & inert. Extends "real assets."
- **taxClass on income streams** — ordinary / tax-free / cap-gains; today all other income is flat ordinary. An engine field, not a label.
- **Sustainable-withdrawal solve** — back into a safe annual portfolio draw, framed as "sustainable withdrawal," not income.
- **Income-stream COLA** — variable income is flat-real today; let fixed annuities erode.
- **Rolling-period analysis** (extended block bootstrap) — sweep every real contiguous historical window, not just the named Sequencing years.
- **Goals page — interesting layout** (not the ledger style). End-date windows already built; the visual redesign is pinned.

7 · Future surfaces / views

- **Comprehensive plan view** (read-only) — the whole plan as a presentation-grade client deliverable. Edits stay on input pages.
- **Prospect-level view** — a lighter "quick illustration" tier for prospects/sales (a reduced view of the same engine).
- **Investment comparisons** — proposed-vs-current / efficient-frontier. $\triangle$ overlaps Scenarios — pin the distinct angle first.

8 · Smaller wins / hygiene

- **Rename "Net Worth" tab** → "Plan" / "Household" (now holds cash flow / goals / snapshot). Quick win.
- **Codify visual grammar** — dashed underline = editable; solid = result; brass = selected/caution; clay = pressure; teal = baseline. Write it down.
- **Assumption ledger** — Entered / Derived / Simulated / Historical tags. $\triangle$ must NOT become the rejected "needs-review" machinery.

9 · Open decisions

- **Demo mortgage** — keep (baseline 84.9%) vs strip to a paid-off house (~86.5%)?
- **Net Worth layout** — make Assets the wider column (Liabilities is often sparse)?
- **SS framing** — lifetime-dollars trade-off vs the % circle (weak % mover for high-spend households).
- **Solve-For display** — solo-vs-combined so an advisor can't misread the column %.
- **Liabilities** — base-plan input vs per-scenario construct.
- **Strategy Fork** — Sequencing feature or Scenarios feature?

10 · Skip / do-not-relitigate logged on purpose

- Cash-flow River / Sankey — decorative, no new interaction.
- Resilience Matrix as a new tab — old Stress Test reincarnated; salvage only the click-to-clone link.
- Rebrand away from "Parallax" — business decision, not engineering.
- Reversed-sequence view — killed (a timeline that never happened).
- Allocation as a solve lever — re-add only with deliberate framing.

Standing doctrine. Make requirements less dumb → delete before adding → simplify → cut beats refactor. The engine is the single truth source; the UI only changes views/levers, never the math. Green is allowed — no mint. Mock visual changes first. Verify by looking. Push to both the working branch and main.